Birla Institute of Technology and Science Pilani (Rajasthan)
EEE F111, ELECTRICAL SCIENCES
First Semester 2025-26
Comprehensive Examination (Close Book-Regular)

Date : 13/12/2025 (AN) **Duration: 3 Hrs.** **Max. Marks: 135**

Note: ***Attempt all parts of particular question at one place (in sequence). Encircle the final answers.***

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Q1:

(A) Given the circuit shown in Fig. 1, find i(t) and v(t) for t > 0 i.e. after the switch is closed. (First calculate the initial conditions, type of response and then proceed). **[15 M]**

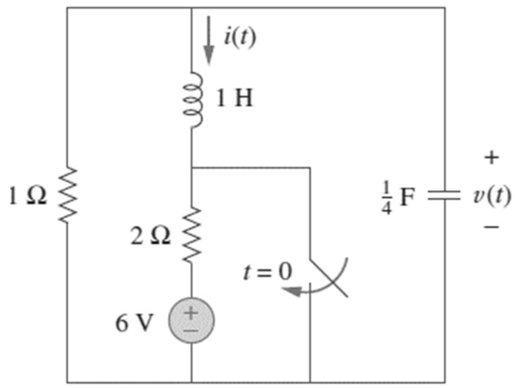

Fig. 1

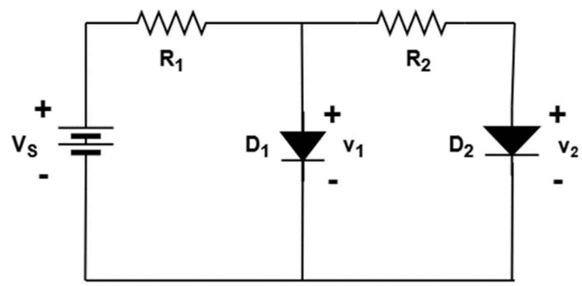

Fig. 2

(B) Suppose that a filter has gain $\left|\mathbf{H}(j\omega)\right| = \dfrac{16\omega}{\left(\omega^2+4\right)}$. (i) For what value of ω is the gain maximum? (ii) What is the maximum value of the gain? (iii) What are the half-power frequencies? (iv) What is the bandwidth of the filter? **[3 + 2 + 3 + 2 = 10M]**

Q2:

(A) For the circuit shown in Fig. 2, V_S = 8V and both the silicon diodes have a saturation current of 1 nA at 300 K.

(a) Find R_2 for the case that R_1 = 20 kΩ and v_1 = 0.66 V.

(b) Find R_1 for the case that R_2 = 200 Ω and v_2 = 0.66 V.

(c) Find R_1 and R_2 for the case that v_1 = 0.68 V and v_2 = 0.66 V.

[15 M]

(B) The silicon BJT in the circuit shown in Fig. 3, has $\beta = 100$. Suppose that $R_1 = 30\ k\Omega$, $R_2 = 20\ k\Omega$, $R_C = 3\ k\Omega$, and $i_C = 2.6\ mA$. Determine the following:

(a) Draw the equivalent circuit diagram showing base resistance R_B in the common-emitter configuration

(b) R_E

(c)) V_{CE} and region of operation

[15 M]

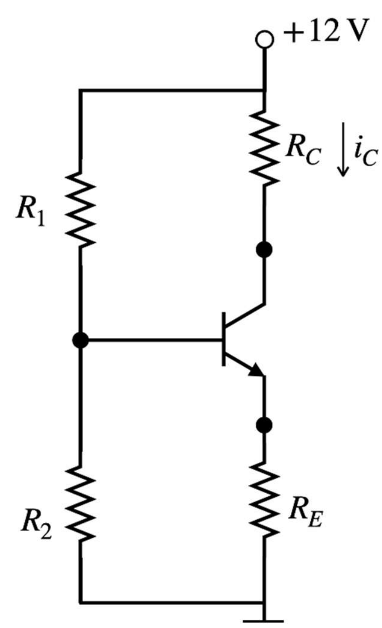

Fig. 3

Q3:

(A) For the circuit shown in Fig. 4, M1 and M2 MOSFETs have I_{DSS} = 1 mA, V_{GS1} = 2 V, R_1 = 1000 KΩ, R_2 = 800 KΩ and $\mathbf{V_P = -1\ V}$, V_{DD} = 18 V. Assuming both the MOSFETS are in active region find (i) i_D (ii) V_{GS2}, V_{GD2} and V_{DS2} (iii) V_{DS1}. Also confirm the region of operation of the MOSFETs M1 and M2 with justification. **[15 M]**

(B) For the ideal OPAMP circuit shown in Fig. 5 find expression for $\mathbf{V_C}$ and $\mathbf{V_O}$, in terms of inputs $\mathbf{V_A}$, $\mathbf{V_B}$, $\mathbf{V_D}$, **and** $\mathbf{V_E}$. (show all steps) **[10 M]**

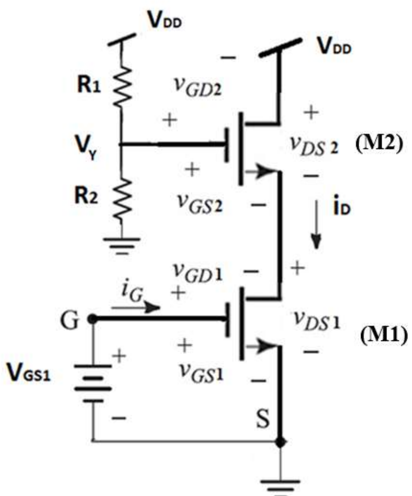

Fig. 4

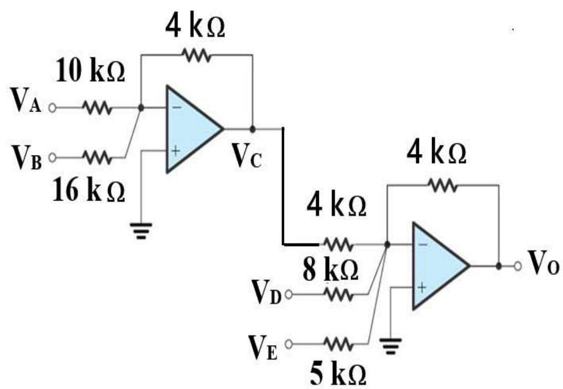

Fig. 5

Q4:

(A) A 50 Hz, three-phase Y-connected 400 V (rms-line) balanced source supplies a Δ-connected three phase balanced load at power factor of 0.8 lagging. The power drawn by per-phase load (Z) is 2 kVA. Calculate:

(a) Per-phase load parameters (R and X)
(b) RMS phase and line current of load
(c) Three-phase active and reactive power drawn by load
(d) Load power factor if supply frequency is changed to 60 Hz **[16 M]**

(B) A magnetic iron core with a relative permeability of 2000 is as shown in Fig. 6. The core has an air-gap of 2 mm. (Given: $\mu_o = 4\pi \times 10^{-7}$ Wb/A-m, all other parameters are depicted on figure itself). Calculate:

(a) Reluctances of magnetic core, air-gap (without fringing effect) and overall magnetic circuit
(b) Number of turns (N) required in coil to produce a flux of 2 mWb
(c) Draw a properly labelled neat and clean electrical analogous circuit diagram.

[15 M]

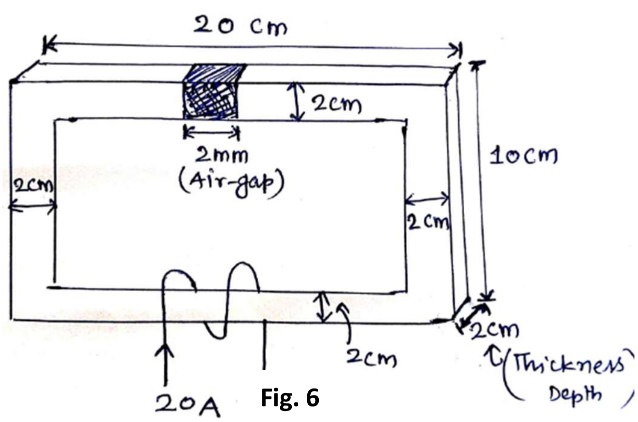

Fig. 6

Q5:

(a) Calculate the phase difference between $v_1 = -10\cos\left(\omega t + 50^0\right)$ and $v_2 = 12\sin\left(\omega t - 10^0\right)$. State which sinusoid is leading. **[3 M]**

(b) For the circuit shown in Fig. 7, find v_0 when 'i_s' is 30 A. **[3 M]**

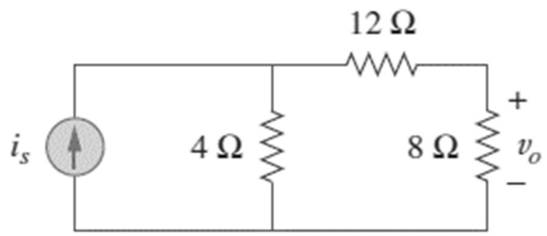

Fig. 7

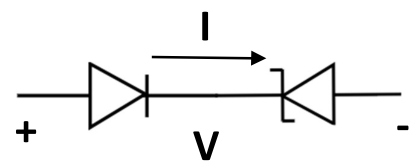

Fig. 8

(c) Consider the diode circuit as shown in Fig. 8. Sketch the I-V characteristics for a voltage sweep of -7 V to +7 V. Assume the reverse breakdown voltage of the Zener diode and conventional PN diode is 6 V and 24 V, respectively. Also consider a 0.7 V voltage drop across both diodes when current flows in the forward direction and zero ON resistance. **[3 M]**

(d) For solving the following question make the following table in your **answer-sheet** and write only the final answers with proper units:

Question Number	Answer
(i)	
(ii)	
(iii)	
(iv)	
(v)	

(For the rough work of these questions use the last page of your answer-sheet or last page of supplement attached)

(i) The forward current through a silicon-based PN junction diode is measured to be 5 mA for a particular Q-point at 300 K. Consider the emission coefficient equal to 2, and compute the dynamic resistance of the diode in Ω. **[3 M]**

(ii) For a NPN BJT, $\alpha = 0.98$ and collector-base junction reverse bias saturation current (I_{CO}) = 0.6 μA. This BJT is connected in the common emitter mode and operated in the active region with a base drive current (I_B) = 30 μA. Find the accurate value of collector current (considering the reverse bias saturation current) for this mode of operation **in μA** . **[3 M]**

(iii) The drain current of an n-channel enhancement mode MOSFET drain is shorted to the gate so that $V_{GS} = V_{DS}$. The threshold voltage of the MOSFET is 1V. If the drain current is 1mA for V_{GS} = 2V then for V_{GS} = 5 V find the value of drain current I_D **in mA**. **[3 M]**

(iv) An ideal single-phase ac source of 50 Hz, 230 V (rms) supplies 20 A (rms) to a load at power factor of 0.7 lagging. Write the load impedance in **polar form**. **[3 M]**

(v) The B-H curve of a magnetic core has value of flux-intensity of 600 AT/m corresponding to flux-intensity of 0.7 T. Calculate value of **relative permeability** of core. **[3 M]**

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